

PRODUCT OVERVIEW · 2026

AnimeAI

A local AI anime studio. Original characters, consistent in every shot, designed and animated fully offline on an 8 GB gaming laptop.

- 100% offline
- RTX 4060 Laptop · 8 GB
- Open-source models
- Scriptable pipeline

18 s	46 min	5.8 GB	$0
per anime keyframe	to teach a new character	peak VRAM in training	per image, forever

Md Zahidul Haque

github.com/zahiddiff/anime-ai

01 · OVERVIEW

From a story idea to anime scenes, on one laptop

AnimeAI chains open-source image, video, language and audio models into one production pipeline for 2D Japanese-style anime in a modern shōnen action look: bold lineart, cel shading, dramatic lighting, dynamic camera.

The problem

- Cloud generators charge per image or clip and keep your work on their servers.
- Generic models draw a **different person every time**, so a character can't carry a story.
- Most local guides assume a 24 GB desktop GPU.

The approach

- Everything runs locally: no accounts, no fees, no upload.
- Each main character gets a small trained add-on (LoRA) that **locks in their look**.
- Every step is tuned and benchmarked for an 8 GB laptop GPU.

01

Character design

Original characters, turnaround sheets and background plates from text.

WORKING

02

Consistency

Train a character once in ~46 min; one trigger word brings them back in any scene.

WORKING

03

Pose control

Copy the exact pose of any reference drawing onto any character.

WORKING

04

Refine & upscale

Anime-specific 4× upscaling plus a detail pass for clean final keyframes.

WORKING

05

Animation

Keyframes become 2–5 s clips, then are interpolated and upscaled.

NEXT

06

Story, voice, edit

Local LLM storyboards, voices, music, SFX and an automatic rough cut.

PLANNED

02 · HOW IT WORKS

One pipeline, eight stages

Every stage is driven through ComfyUI's HTTP API by small Python scripts. Workflows are generated in code and stored as JSON, so a batch job can fill in prompts, seeds and sizes and render hundreds of shots unattended.

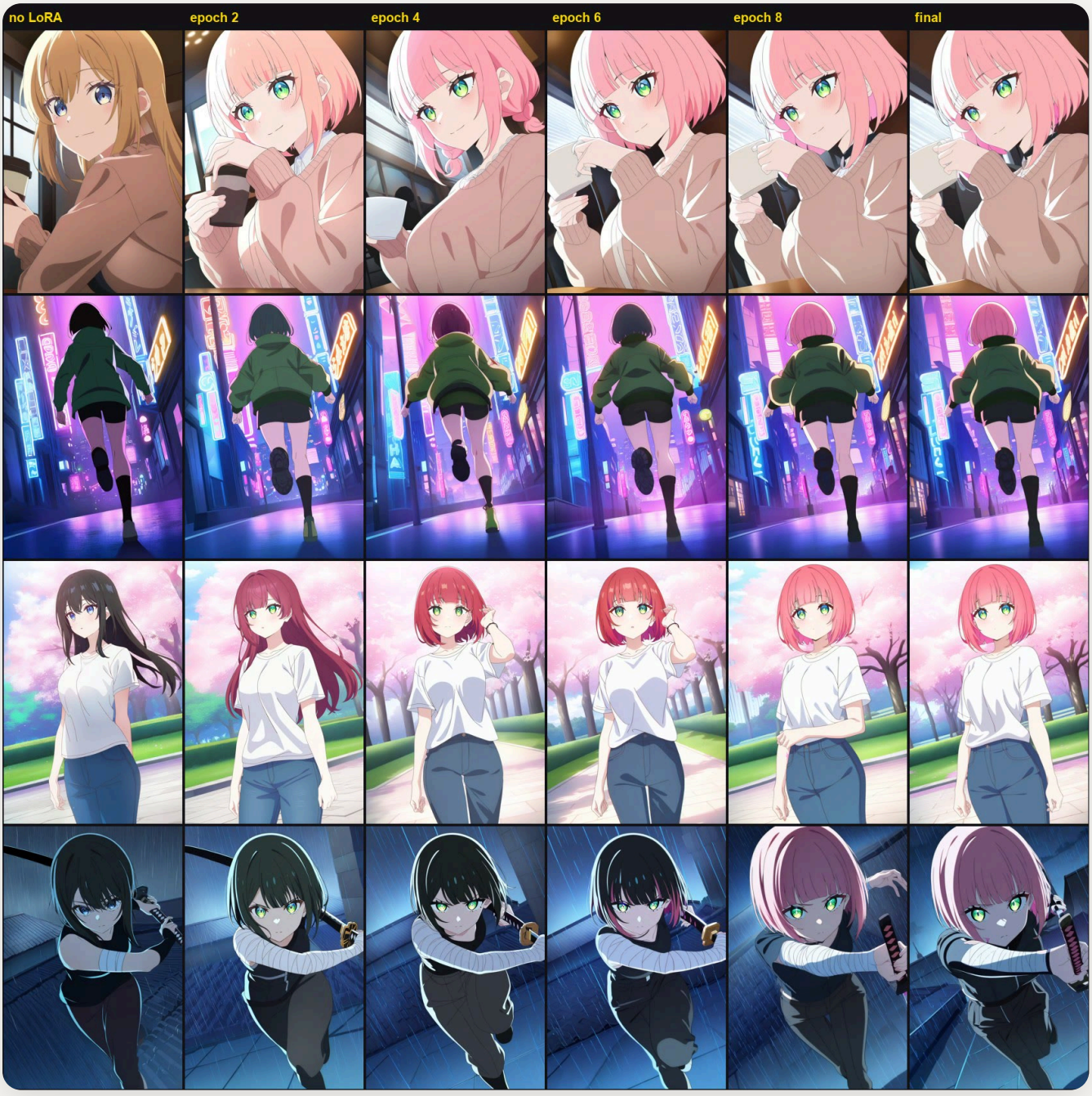
01	**Story → shot list** A local language model turns a premise into a short script and a machine-readable shot list: camera, action, dialogue, prompts.	Ollama · Qwen / Gemma
02	**Character design** Concepts, turnaround sheets and backgrounds from the NoobAI-XL anime model.	NoobAI-XL v1.1
03	**Character training** Generate 48 varied images → curate → auto-caption → train an 8 GB-safe LoRA.	kohya sd-scripts · WD14
04	**Keyframes** Character LoRA + ControlNet pose for each shot's key moment.	ControlNet Union ProMax
05	**Refine & upscale** 4× anime upscaler, then a low-denoise pass for crisp lines.	AnimeSharp · RealESRGAN
06	**Image → video** Animate keyframes into short clips; first + last frame control for action beats.	Wan 2.2 (5B / 14B GGUF)
07	**Voice, music, SFX** Local text-to-speech, music generation and video-to-audio effects.	GPT-SoVITS · ACE-Step
08	**Assemble** RIFE interpolation, then an ffmpeg rough cut exported for DaVinci Resolve.	RIFE · ffmpeg

Working Next Planned

03 · CHARACTER CONSISTENCY

Same character, every scene

A base model draws a new person each time. After 46 minutes of training on 37 curated images, the trigger word **sumi_ink** brings back the same character in scenes she was never trained on, while her clothes still follow the prompt.



Rows: café · neon-city run · casual park · rainy sword fight. Columns: no LoRA → training epochs 2, 4, 6, 8 → final. The prompts never mention hair or eye colour.

Without LoRA

Four different people: blonde, black-haired, brunette.

Epochs 2–6

Learning, but unstable; hair drifts to red or black in dark scenes.

Epoch 8 → final

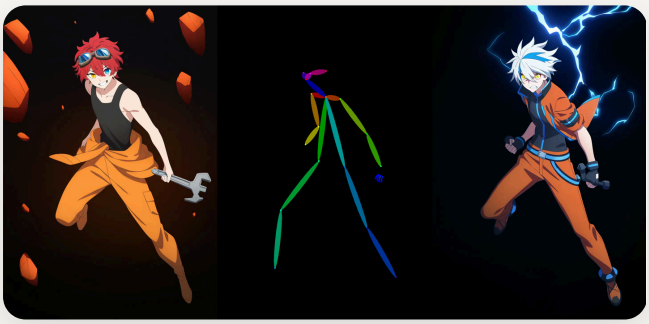
Pink bob, blunt bangs, green eyes, reliably, in all four scenes.

04 · DESIGN TOOLS

Concepts, poses, sheets, backgrounds



Three original characters designed from text: storm courier, ink swordswoman, gravity mechanic.



Pose control: reference → extracted skeleton → a different character in the same pose.



Refine + upscale: 832×1216 → 1248×1824 with cleaner lines.



Turnaround reference sheet.



Background plate, no characters.

05 · PERFORMANCE

Built for an 8 GB laptop

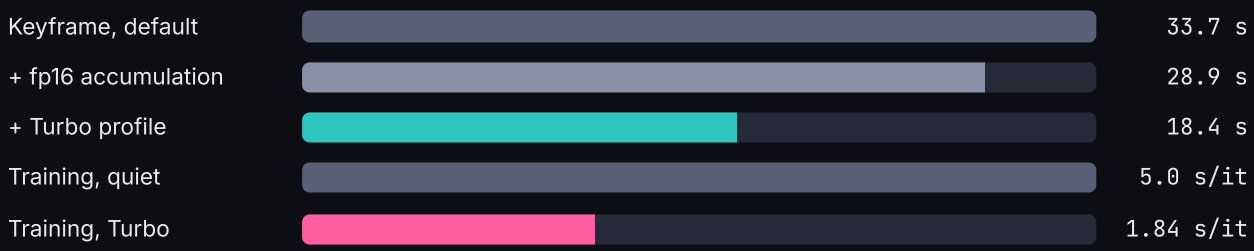
Measured on an Acer Nitro V 16: RTX 4060 Laptop (8 GB), Core i5-14450HX, 16 GB single-channel DDR5.

TASK	OUTPUT	TIME	PEAK VRAM
Anime keyframe	832×1216, 28 steps	**18 s**	7.5 GB
Pose-controlled keyframe	OpenPose + ControlNet	**51 s**	7.3 GB
Refine + 4× upscale	→ 1248×1824	**55 s**	7.7 GB
Character LoRA training	1,480 steps, rank 16	**46 min**	5.8 GB

TUNING FINDING

The laptop was throttling itself

The vendor's quiet fan profile capped the GPU at 1.2 GHz and ~20 W at only 58 °C. Switching to the Turbo profile raised it to 2.4 GHz and up to 91 W.



What fits in 8 GB

fp8 base weights, UNet-only LoRA training, cached latents and text-encoder outputs, tiled VAE decode, dynamic VRAM loading.

What doesn't

`--highvram` with ControlNet exceeds 8 GB and spills into shared memory: 225 s per step instead of ~1 s. Avoided.

06 · TECHNOLOGY

Open-source stack

LAYER	TOOL	LICENCE
Engine	ComfyUI v0.37 portable · PyTorch 2.13 · CUDA 13	GPL-3.0
Anime model	NoobAI-XL v1.1 (SDXL / Illustrious family)	FAIPL-1.0-SD
Pose / structure	xinsir ControlNet Union ProMax · OpenPose	Apache-2.0
Upscaling	4x-AnimeSharp · RealESRGAN x4plus anime	CC BY-NC-SA · BSD-3
Training	kohya-ss sd-scripts v0.12 · WD14 tagger v3	Apache-2.0
Video (next)	Wan 2.2 TI2V-5B / I2V-14B GGUF · RIFE	Apache-2.0 · MIT
Automation	Python · ComfyUI HTTP API · ffmpeg	MIT (this project)

07 · ROADMAP

What's next

Done

- Hardware audit & tuning
- ComfyUI engine
- Character, pose, sheet, background workflows
- 8 GB character LoRA training

Next

- Image → video (Wan 2.2 5B vs 14B)
- First + last frame control
- RIFE interpolation + upscale
- Sword-draw action test shot

Planned

- LLM storyboard → shot list
- Voices, music, SFX
- Overnight batch renderer
- 15–20 s end-to-end scene

Responsible use

Original characters only; no models that reproduce existing anime characters. Model licences are respected: FAIPL derivatives stay open, and the commercial-safe upscaler is used for monetised work. Voice cloning only with voices the author has rights to.



ABOUT

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AI & Computer Engineering student at Tongmyong University, Busan. Builds end-to-end systems across AI pipelines, e-commerce and desktop apps.

Project	github.com/zahiddiff/anime-ai
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